

Aim:

Write a Python program using the NumPy library to compute the following statistical measures for a given set of numbers:

Central Tendency Measures

- Mean
- Median
- Mode

Measure of Dispersion

- Population Variance (formatted to two decimal places)
- Population Standard Deviation (formatted to two decimal places)

The program should take the elements of the array as space-separated integers from the user and display the required statistical measures in the specified format.

Input Format:

- A single line containing space-separated integers representing the elements of the array.

Output Format:

- Display the following five values, each on a separate line in the exact order:

```
Mean: <mean value>
Median: <median value>
Mode: <mode value>
Variance: <variance value formatted to 2 decimal places>
Standard Deviation: <standard deviation value formatted to 2 decimal places>
```

Notes:

- Compute the population variance and population standard deviation using `np.var()` and `np.std()`. These functions compute the population values by default
- If multiple modes exist, print the mode with the highest frequency that appears first in the sorted array.

Source Code:

StatisticalMeasures.py

```
# Type Content here...
import numpy as np
arr=np.array(list(map(int,input().split()))))

mean=np.mean(arr)
median=np.median(arr)

sorted_arr=np.sort(arr)
values,counts=np.unique(sorted_arr,return_counts=True)
mode=values[np.argmax(counts)]

variance=np.var(arr)
std_dev=np.std(arr)
```

```
print("Mean:",mean)
print("Median:",median)
print("Mode:",mode)
print(f"Variance: {variance:.2f}")
print(f"Standard Deviation: {std_dev:.2f}")
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
1 2 3 4 4 5 5 5
Mean: 3.625
Median: 4.0
Mode: 5
Variance: 1.98
Standard Deviation: 1.41

Test Case - 2
User Output
-5 -3 -1 0 2 4 4 4
Mean: 0.625
Median: 1.0
Mode: 4
Variance: 10.48
Standard Deviation: 3.24